

Design a database for YB college:

1. Project scope (short paragraph)

Write a story that defines the purpose and scope of the database. Describe the main entities (e.g., students, lecturers, etc).

A database that tracks course enrollments and allows access for students, lecturers, and admins. This database will track the number of enrollees per course. It will allow students and lecturers to access relevant data. Lastly, this database will enable admin users to manage course data. Here's a breakdown of the **entities**, their **attributes**, and the **relationships** between them:

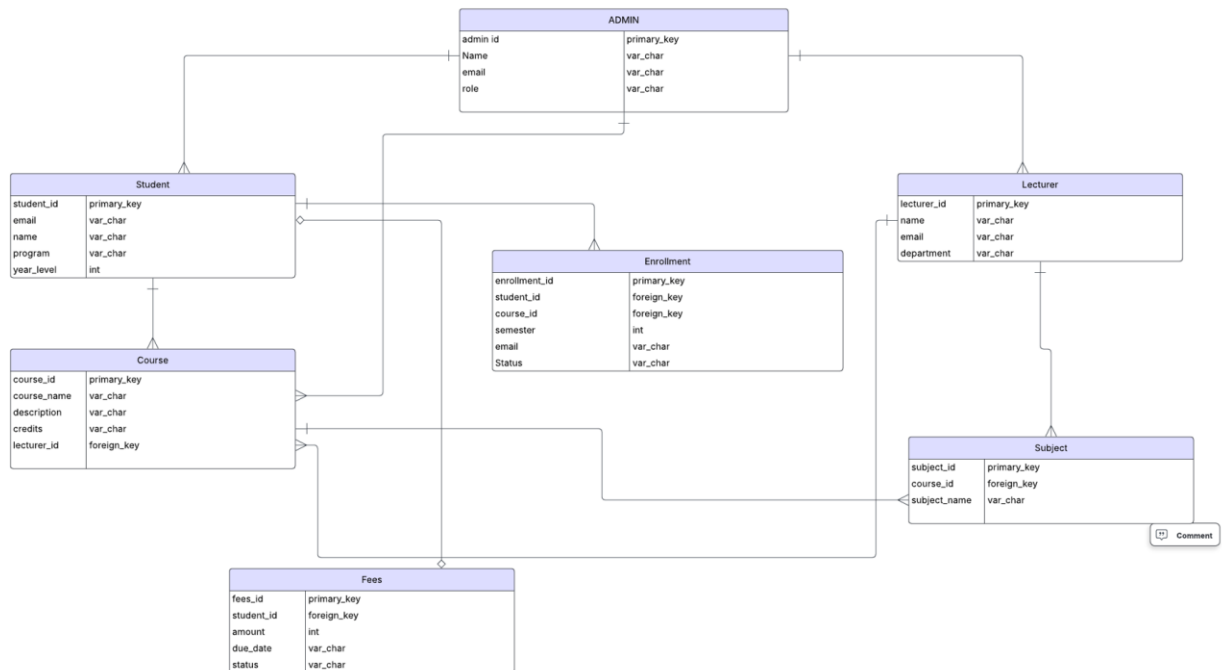
- A. Student
- B. Lecturer
- C. Admin
- D. Course
- E. Enrollment
- F. Fees
- G. Subjects

Relationships

- **Student ↔ Enrollment ↔ Course**
A student can enroll in multiple courses; each course can have many students.
- **Lecturer ↔ Course**
A lecturer teaches one or more courses.
- **Course ↔ Subject**
A course can have multiple subjects.
- **Student ↔ Fees**
Each student has associated fee records.
- **Admin ↔ Student / Lecturer / Course**
Admin can add, update, or delete records for students, lecturers, and courses.

2. Entities and EER diagram

List all entities with brief descriptions of their roles and attributes (e.g., Student, Course, Class, Lecturer, etc).



3. Table design

State how many tables are required after mapping the EER to a relational schema.

1. Student

student_id (Primary Key)

name

email

program

year_level

2. Lecturer

lecturer_id (Primary Key)

name

email

department

3. Admin

admin_id (Primary Key)

name

email

role

4. **Course**
 - course_id (Primary Key)**
 - course_name**
 - description**
 - credits**
 - lecturer_id (Foreign Key)**
5. **Subject**
 - subject_id (Primary Key)**
 - subject_name**
 - course_id (Foreign Key)**
6. **Enrollment**
 - enrollment_id (Primary Key)**
 - student_id (Foreign Key)**
 - course_id (Foreign Key)**
 - semester**
 - status (e.g., enrolled, dropped, completed)**
7. **Fees**
 - fee_id (Primary Key)**
 - student_id (Foreign Key)**
 - amount**
 - due_date**
 - status (paid/unpaid)**